

GOVERNMENT ENGINEERING COLLEGE, SHEOHAR

Department of Electronics & Communication Engineering

Team:-ROBOSENSE

Project Name:-Bluetooth Controlled RC Car & IoT-Based Smart Multi-Gas Environment Monitoring System

A Dual-Purpose Mobile Robotic Platform for Wireless Control and Gas Monitoring

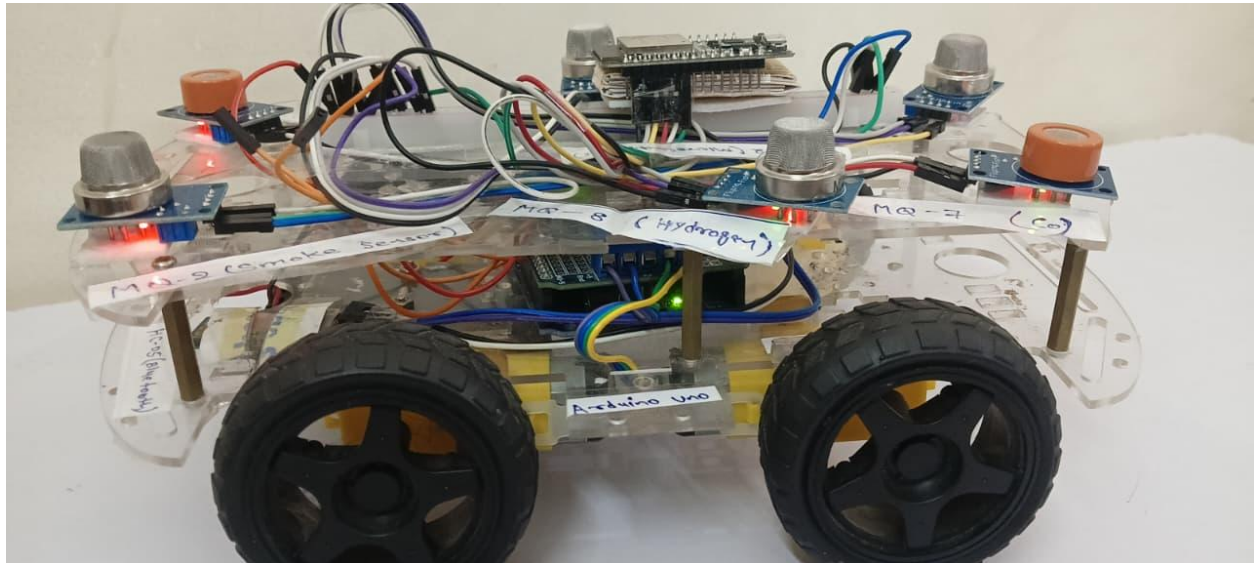


Figure 1. RoboSense – assembled prototype

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Institution	Government Engineering College, Sheohar, Bihar

1. Project at a Glance

Project Name: RoboSense

What was the project? RoboSense is a two-part embedded-systems project built on a common four-wheel acrylic robotic chassis. In the first mode, the chassis works as a Bluetooth-controlled RC car using an Arduino Uno, HC-05 Bluetooth module and L293D-based motor driver. In the second mode, the same chassis becomes an IoT-based smart environment and multi-gas monitoring robot using an ESP32, six MQ-series gas sensors and the Blynk IoT cloud platform.

Core idea: combine mobile robotic movement with environmental sensing so that the robot can be moved to different locations while monitoring hazardous gases and sending sensor readings and alerts to a mobile dashboard.

Project Description

The project demonstrates two complementary electronics and embedded-systems applications on one robotic platform. The RC-car section provides wireless mobility through Bluetooth. A smartphone sends simple direction commands such as F, B, L and R to the HC-05 module, and the Arduino Uno controls four BO gear motors through the L293D motor driver shield. The second section uses an ESP32 to read six MQ-series gas sensors: MQ-2, MQ-3, MQ-4, MQ-7, MQ-8 and MQ-9. Their analog readings are sent through Wi-Fi to Blynk, where they can be viewed on a mobile dashboard. When a sensor reading crosses the fixed danger threshold used in the project, the system activates the warning LED and generates a Blynk Cloud danger-alert notification.



Figure 2. Side view of the RoboSense prototype

Main Features

- Four-wheel robotic chassis using four BO gear motors and rubber wheels.
- Bluetooth-based wireless movement using an HC-05 module and Arduino Uno.
- L293D motor driver shield for independent control of four motor channels.
- ESP32-based monitoring section with six MQ-series gas sensors.
- Wi-Fi connectivity and real-time sensor display through the Blynk IoT platform.
- Danger alert using a Blynk Cloud event and a local LED indicator.
- Common mechanical chassis allows both subsystems to be demonstrated on one platform.

2. Objectives

- To design and assemble a four-wheel-drive robotic chassis using BO motors and an L293D motor driver.
- To control the chassis wirelessly using an HC-05 Bluetooth module interfaced with an Arduino Uno.
- To interface six different MQ-series gas sensors (MQ-2, MQ-3, MQ-4, MQ-7, MQ-8 and MQ-9) with an ESP32.
- To read analog gas-sensor values and transmit them to the Blynk IoT cloud dashboard over Wi-Fi.
- To generate a real-time danger alert through a mobile notification and onboard LED when a sensor crosses the fixed safety threshold.
- To document the hardware, wiring, firmware and working of both subsystems in one project.

3. Major Components Used

Component	Function in RoboSense
Arduino Uno	Main microcontroller for the Bluetooth-controlled RC-car mode.
HC-05 Bluetooth Module	Wireless serial link between smartphone and Arduino Uno.
L293D Motor Driver Shield	Drives the four BO motors and provides H-bridge motor control.
BO Gear Motors ×4	Provide mechanical drive to the four wheels.
Rubber Wheels ×4	Provide traction and mobility.
18650 Li-ion Battery Pack	Rechargeable power source for the robotic platform.
ESP32 Dev Board	Wi-Fi-enabled microcontroller for gas sensing and Blynk communication.
MQ-2	Smoke and combustible-gas sensing.
MQ-3	Alcohol-vapour sensing.
MQ-4	Methane / natural-gas / CNG sensing.
MQ-7	Carbon-monoxide (CO) sensing.
MQ-8	Hydrogen (H ₂) sensing.
MQ-9	Carbon monoxide and combustible/flammable-gas sensing.
Warning LED	Local visual indication when the danger condition is detected.

4. Working Principle

4.1 Bluetooth Controlled RC Car

An Android smartphone is paired with the HC-05 Bluetooth module. A Bluetooth terminal or RC-car application sends a single character for each direction: F for forward, B for backward, L for left and R for right. The Arduino Uno receives these characters through SoftwareSerial and first stops the motors before executing the new command. For forward and backward movement, all four motors rotate in the required direction. For turning, motors on opposite sides are driven in opposite directions, producing a differential/skid-steer turn.

4.2 IoT-Based Smart Multi-Gas Monitoring

In monitoring mode, the ESP32 reads the analog outputs of six MQ-series sensors. The sensors are connected to ESP32 ADC1 pins so that analog sensing can operate together with Wi-Fi. The ESP32 ADC provides 12-bit readings from 0 to 4095. The readings are uploaded to Blynk virtual pins V2 to V7 every four seconds and displayed on the mobile dashboard using gauge/chart widgets.

The firmware compares all six readings with a common GAS_THRESHOLD value of 2800. If any reading becomes greater than this value, the warning LED is switched to its danger state and the ESP32 can trigger the Blynk Cloud "danger_alert" event. A 60-second interval is used between alerts to reduce repeated notifications.

4.3 Sensor Mapping

Sensor	ESP32 Pin	Blynk Virtual Pin
MQ-2	GPIO 32	V2
MQ-3	GPIO 33	V3
MQ-4	GPIO 34	V4
MQ-7	GPIO 35	V5
MQ-8	GPIO 36	V6
MQ-9	GPIO 39	V7

4.4 Key Parameters Used

- ESP32 ADC resolution used: 12-bit (0–4095).
- Sensor data update interval: every 4 seconds.
- Common gas danger threshold: 2800 ADC counts.
- Minimum interval between danger alerts: 60 seconds.
- Blynk event name: danger_alert.
- Blynk virtual pins: V2–V7.

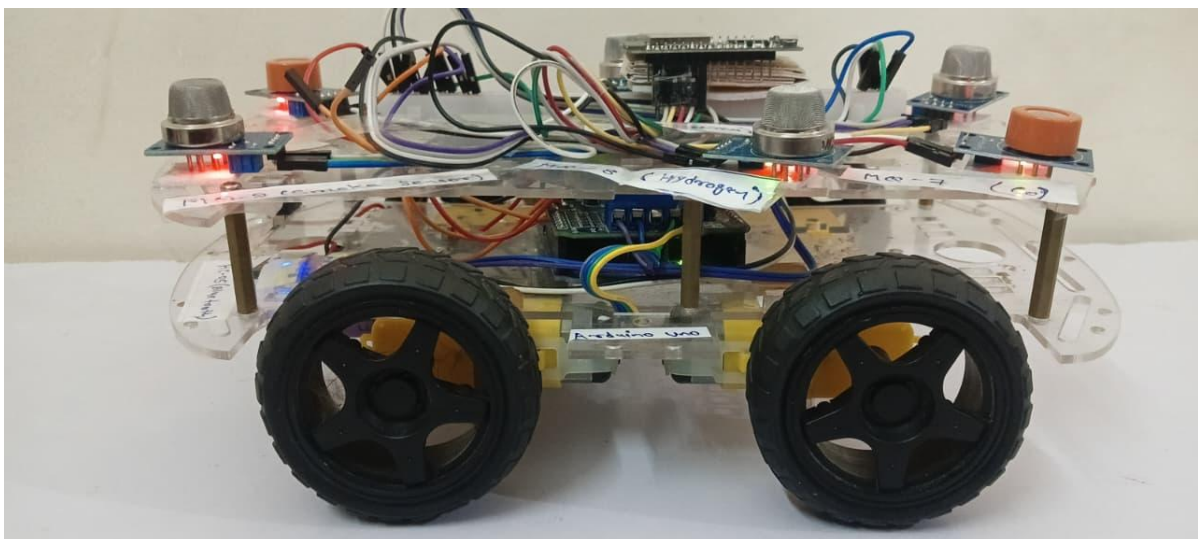


Figure 3. Front/side view showing the labelled gas sensors and chassis

5. Project Photographs

The following photographs show the assembled RoboSense prototype from different views. The chassis contains the four-wheel drive base, Arduino Uno/L293D motor-control section, ESP32 monitoring section, HC-05 Bluetooth module, six MQ-series gas sensors, wiring and power arrangement.

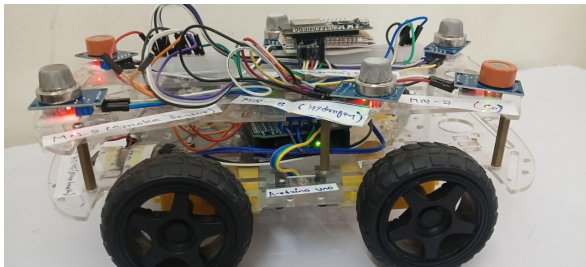


Figure 5.1 – RoboSense prototype view



Figure 5.2 – RoboSense prototype view

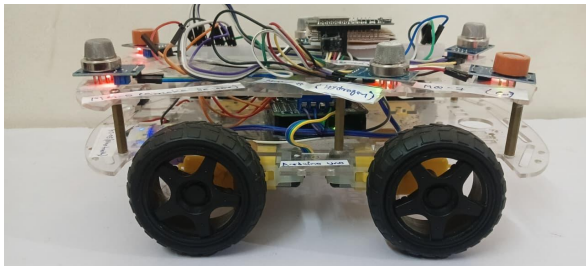


Figure 5.3 – RoboSense prototype view

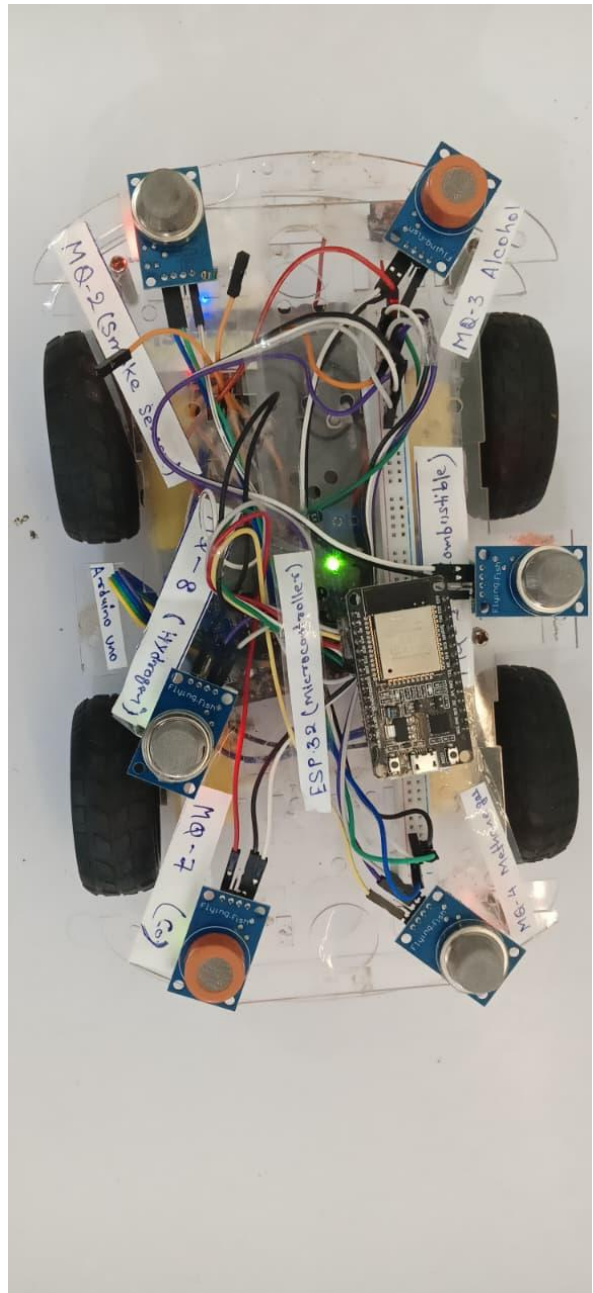


Figure 5.4 – RoboSense prototype view

6. Project Team



Figure 6.1. Project team with the RoboSense project demonstration

The project was prepared and demonstrated by the following team members. The project lead coordinated the overall project work, including the integration and documentation of the two subsystems.

Name	Role	Branch	Registration No.
Shivshankar Kumar	Project Lead	ECE	24104145007
Nitish Kumar	Team Member	EE	24103145010
Mantosh Kumar	Team Member	ECE	23104145008
Ashutosh Kumar Jha	Team Member	CS	25153145901

7. Applications

- Remote-controlled inspection of confined or potentially hazardous spaces.
- Early LPG/methane leak detection in kitchens and gas-storage areas.
- Carbon-monoxide monitoring in enclosed spaces such as generator rooms or parking garages.
- Mobile smoke/fire early-warning patrol applications for safety rounds.
- Educational demonstration of motor control, Bluetooth communication, ADC sensing and IoT cloud integration.

8. Advantages and Limitations

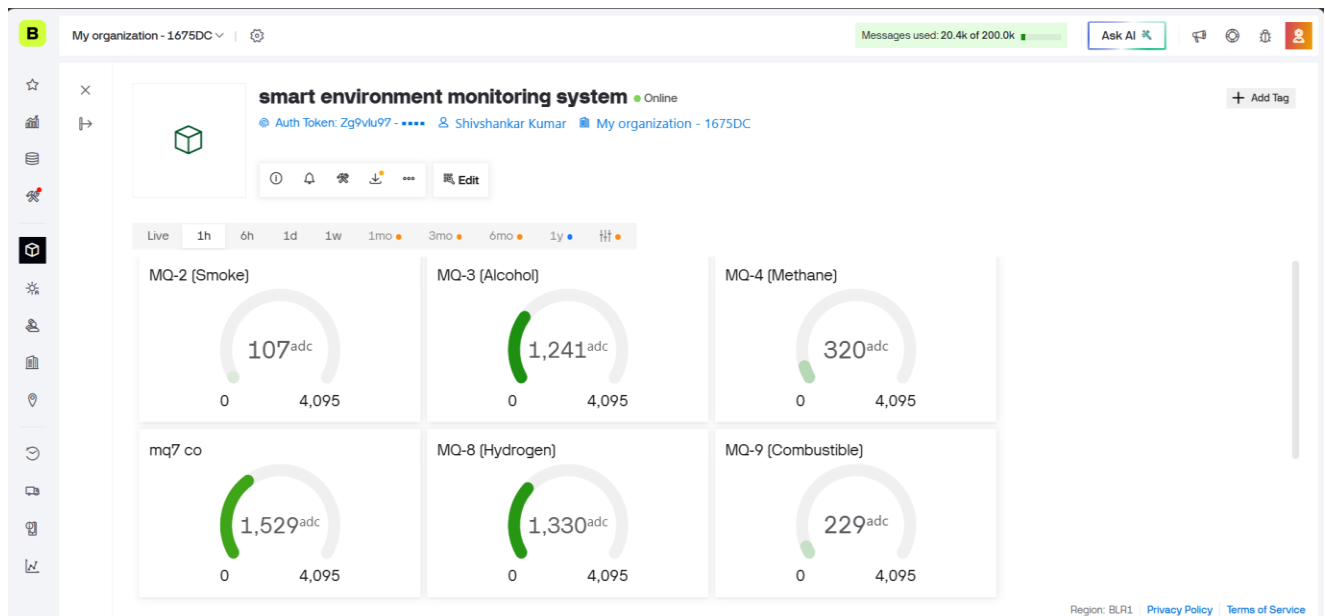
Advantages

- Low-cost components make the prototype suitable for a student project.
- Mobility allows gas readings to be taken at different physical locations.
- Blynk dashboard and push notifications allow remote monitoring.
- The RC-car and gas-monitoring sections are modular and can be used independently.

Limitations

- MQ-series sensors require warm-up/burn-in and periodic calibration for accurate readings.
- Bluetooth and Wi-Fi currently use separate firmware images rather than one simultaneous firmware mode.
- Wireless operating range depends on Bluetooth and Wi-Fi coverage.
- The present firmware uses one common raw-ADC threshold for all six sensors rather than gas-specific calibrated ppm thresholds.

Monitor:-IoT-Based Smart Multi-Gas Environment Monitoring System



9. Future Scope

- Merge both firmware systems so the ESP32 can drive the motors and monitor gases using its Wi-Fi and Bluetooth capabilities.
- Use individual sensor-specific thresholds in ppm after proper calibration.
- Add GPS so danger alerts can include the robot's location.
- Add ultrasonic/IR obstacle-avoidance sensors for semi-autonomous patrolling.
- Store historical sensor data in cloud storage or an external database for trend analysis.

10. Conclusion

RoboSense demonstrates a practical combination of robotics, embedded control, wireless communication and IoT-based environmental sensing on a common robotic chassis. The first subsystem provides Bluetooth-controlled mobility using Arduino Uno, HC-05, L293D motor control and four BO motors. The second subsystem uses ESP32 and six MQ-series sensors to collect gas-related analog readings and send them to the Blynk IoT platform for remote monitoring and danger alerts.

Through the development and testing of the prototype, the team gained hands-on experience in Arduino and ESP32 programming, motor-driver interfacing, Bluetooth and Wi-Fi communication, analog sensor interfacing, cloud dashboards and embedded-system integration. The assembled hardware photographs document the completed prototype and its major modules.

11. Project Summary

Project	RoboSense
Platform	Four-wheel acrylic robotic chassis
RC Control	Arduino Uno + HC-05 Bluetooth + L293D + 4 BO motors
Monitoring	ESP32 + MQ-2, MQ-3, MQ-4, MQ-7, MQ-8, MQ-9
Cloud	Blynk IoT
Data Update	Every 4 seconds
Danger Threshold	2800 / 4095 ADC
Alert	Blynk danger_alert event + local LED

Prepared for Academic / Mini-Project Presentation

Government Engineering College, Sheohar, Bihar